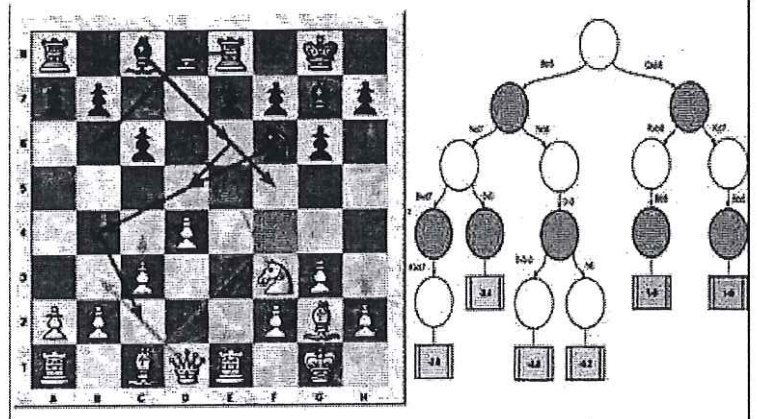


**BITS PILANI DUBAI CAMPUS
INTERNATIONAL ACADEMIC CITY, DUBAI
ACADEMIC-UNDERGRADUATE STUDIES DIVISION
BITS F464- MACHINE LEARNING
FIRST SEMESTER 2019-2010**

DATE: 24.09.2019

1. Having Fun. Hope you identify the game given below. What machine learning algorithm should be opted by the player to win the game? (Gain 0.25 points)



Model complexity [polynomial degrees]	Training Error [MAE]	Testing Error [MAE]
1	8.5	8.5
2	3.5	4.0
3	2.5	2.8
4	2.4	3.0
5	2.3	3.2
6	2.2	3.4
7	2.1	3.8
8	2.0	4.2
9	1.9	5.0
10	1.8	7.5
11	1.7	19.5
12	1.6	24.0

If yes/no, Justify? (Gain 0.25 points)

4. Given below is the polynomial regression equation. If 'D' is the dimensions of the dataset and 'x' is the observation or data points.

Is the regression equation given below is an ideal equation?? (Gain 1 point)

Do you think u can rectify the problem? If so how? (Gain 1 point)

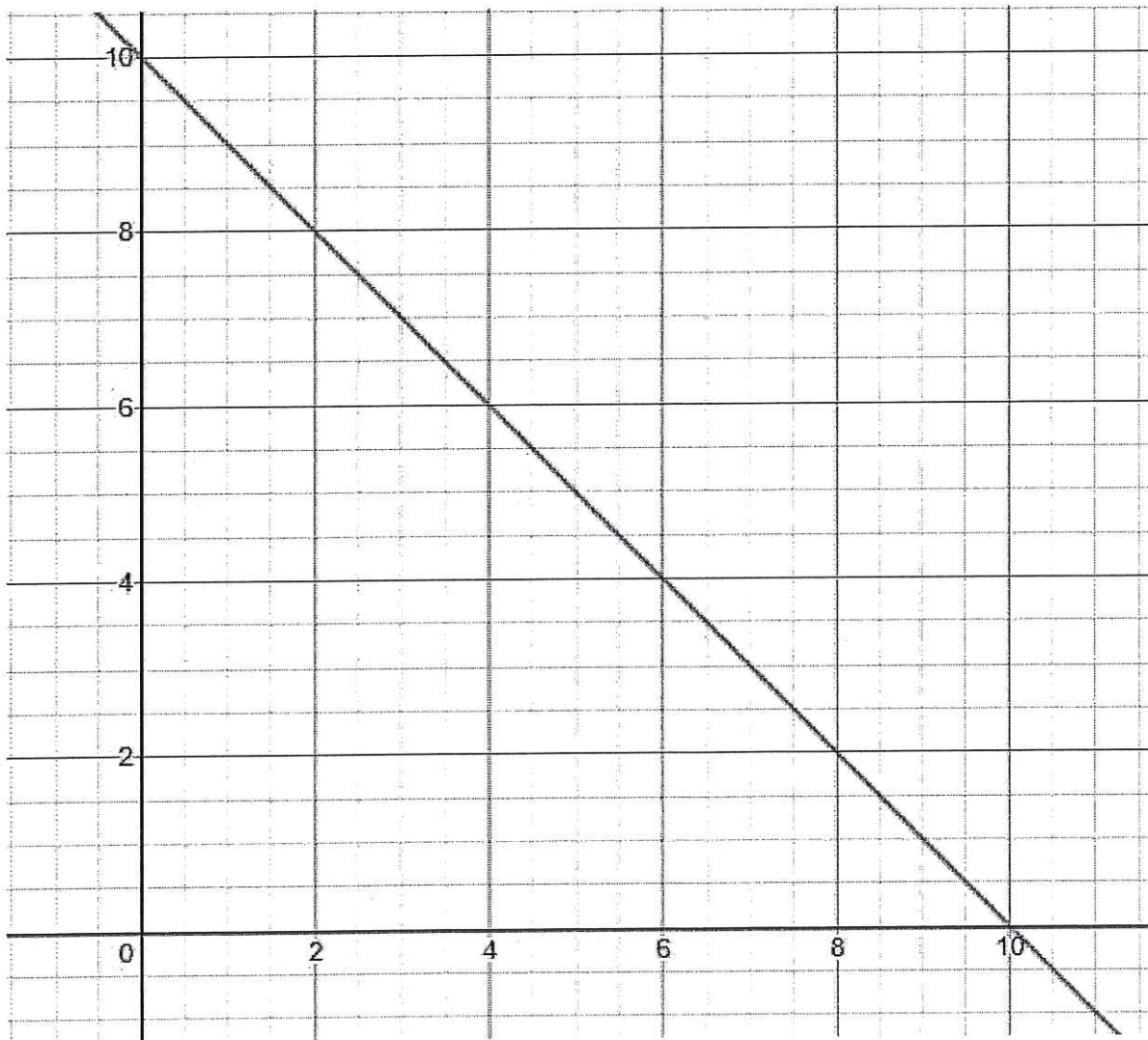
$$y(X, W) = w_0 + \sum_{i=1}^D w_i x_i + \sum_{i=1}^D \sum_{j=1}^D w_{ij} x_i x_j + \sum_{i=1}^D \sum_{j=1}^D \sum_{k=1}^D w_{ijk} x_i x_j x_k + \sum_{i=1}^D \sum_{j=1}^D \sum_{k=1}^D \sum_{l=1}^D w_{ijkl} x_i x_j x_k x_l + \dots + \dots$$

5. Given below is a graph.

Derive the equation of the line. (Gain 2 point)

Assuming the line to be the predicted regression line, calculate the error incurred when the following

test points are subjected to the graph $\{(4,4), (2,6), (4.5,4), (2.25,6.25), (8.25,2.25)\}$ (Gain 3 point)

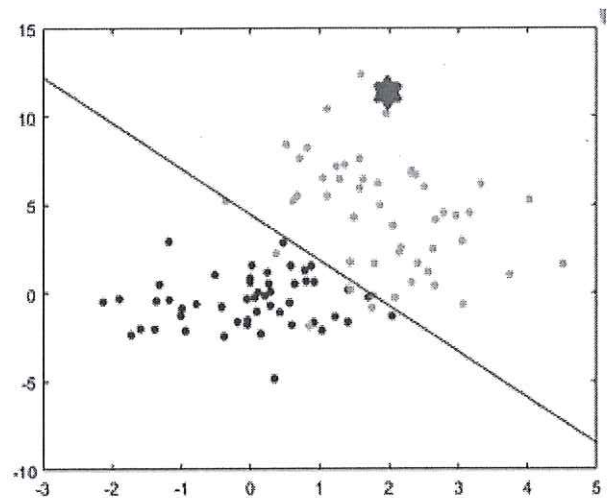


6. Given function classifier be

low:

To which class you will categorize the 'STAR POINT' and how?? Which methodology do you opt? **(Gain 1 points)**

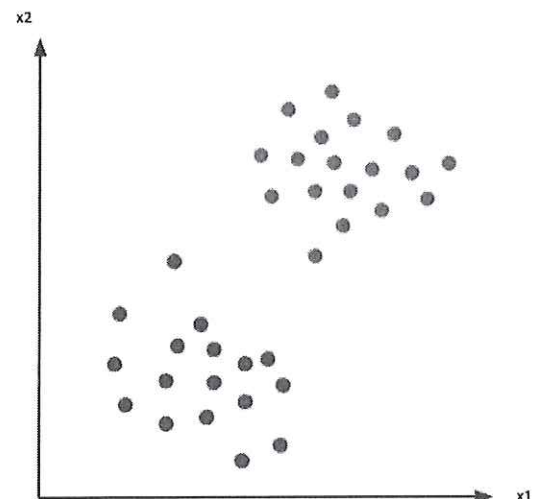
Using the help of Linear Algebra and determine the optimal solution. Draw the geometry of the hypothesis as you solve and derive the solution. **(Gain 5 points)**



7. Given the data points plotted in graph with x_1 and x_2 as two parameters.

Identify an axis over which the feature vectors can be projected so as to minimize the overlapping. Draw the same. **(Gain 1 points)**

Construct a decision boundary considering the intra distance between the feature vectors of each class is very high. Derive and Justify the method adopted to construct the decision boundary. **(Gain 5 points)**



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BITS F464- MACHINE LEARNING – TEST 2
FIRST SEMESTER 2019-2010

TIME: 50 MINUTES
DATE: 12.11.2019

MARKS: 25

1. For overlapping data points, Identify the efficient classifier.

(1 point)

- A. Bayes classifier
- B. Bayes optimal classifier
- C. Support vector machines with linear kernel
- D. None of the above

2. Assume you being a data scientist. You are given with a data set and an efficient algorithm and asked to analyze the prediction behavior of the algorithm in different dimensions. In a generic way, you predict the behavior of the algorithm by applying a number of hypothesis during testing and measure its performance. You end up in a solution that the algorithm behaves in a different fashion against different sets of hypothesis.

And now you are given with a new hypothesis, identify the suitable algorithm you will chose to evaluate the new hypothesis. (1 point)

3. Given:

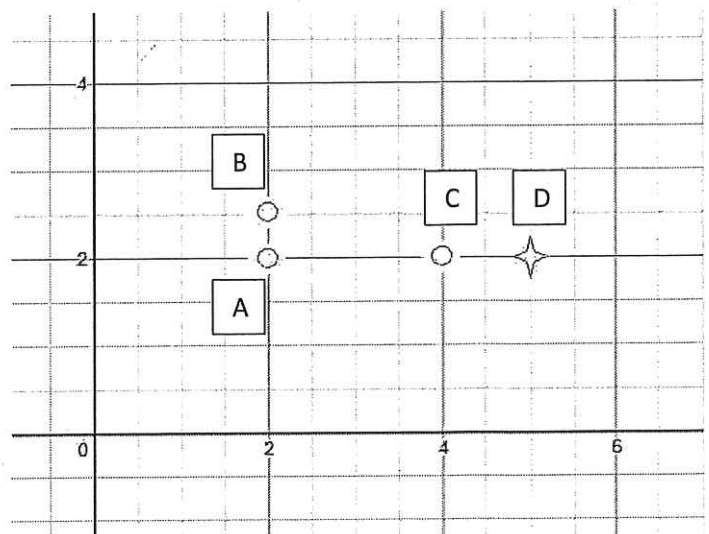
$2x - y = 8$ is the equation of white line on the road.

The road above separate two different cities. Find the width of the road. (2 points)



4. Given the points:

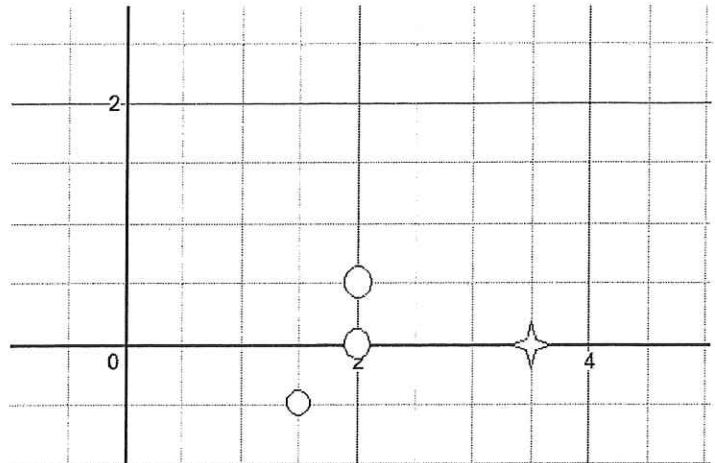
Identify the suitable classifier equation using an Ensemble Learning approach. (5 points)



5. Classify the points in the graph using a hyperplane. It's known that the support vectors need to be identified for classification. Identify the optimal support vectors from the graph and calculate the parameters (α) for the support vectors.

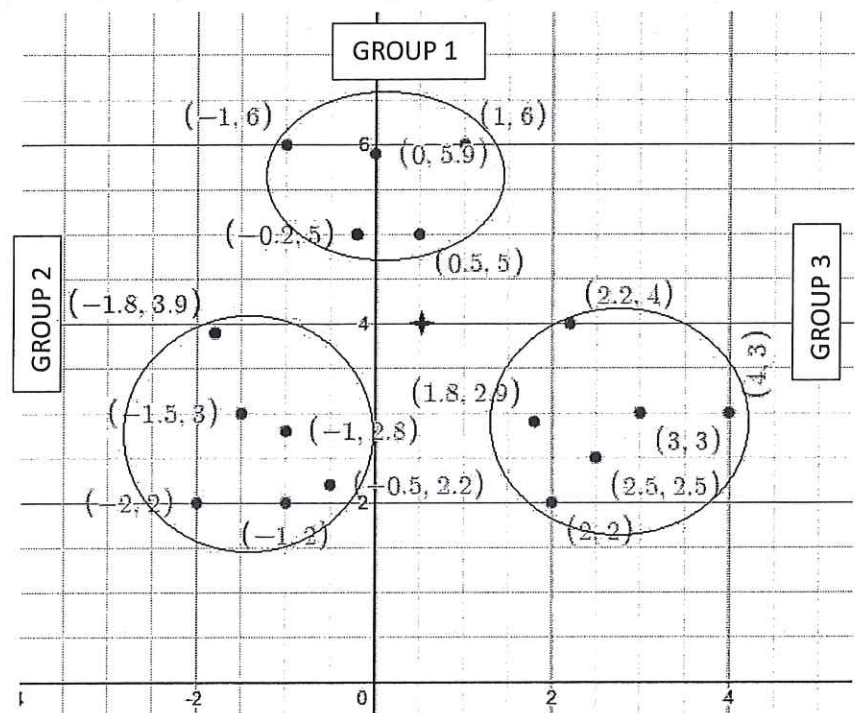
(5 points)

Get the x and y coordinates of the points from the graph.



6. Given three groups of data points. Use 3-NN, a Lazy learner to identify the neighbors for the "star point" and classify it to one of the groups. Use formulae. (5 points).

Get the x and y coordinates of the star point from the graph.



7. Out of 1000 patients, 150 patients were identified with Alzheimer syndrome every year.

Memory loss vs Alzheimer analysis was conducted with patients. It was found that 97% of people with Alzheimer were identified with memory loss. Also 11% of people without Alzheimer were identified with memory loss.

HYPOTHESIS: Does a patient with memory loss can be predicted to have Alzheimer.

Use a Generative algorithm that you are familiar with and solve the hypothesis. Also apply one optimization technique for the chosen algorithm and compare the results. (6 points)

4/5

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BITS F464- MACHINE LEARNING
COMPREHENSIVE EXAMINATION
FIRST SEMESTER 2019-2010

TIME: 180 MINUTES

MARKS: 40

DATE: 23

.12.2019

Structure

QUESTION	POINTS	TIME
1. Identify the good thief from Machine Learning perspective	0.5	5 min
2. Dependency Tree	0.5	5 min
3.a. Gain reward	0.5	10 min
3.b. Think it Through	0.5	
4. Zoologist confused with Tiger and Cheetah	2	10 min
5. Stop before it's too high	2	10 min
6. Let it maximize	4	15 min
7. Drive into and not out	4	15 min
8. Build a strong house with small weak bricks	4	15 min
9. a. Melange of Models	3	15 min
9. b. Burglary Alarm	1	
10. Fine tuning the network but how long???	9	30 min
11. Too much math involved to reduce the weight of anything	9	30 min

1. Identify the good thief from Machine Learning perspective (0.5 points)

Assume there are many houses in a street where different types of people live (living on streets, very poor, poor, moderate, rich, and very rich). One fine day six houses were burgled by a thief.

Analyze the options and choose the way of theft that would maximize the value of entropy

- a. When the thief steals from all rich people.
- b. When the thief steals from all poor people.
- c. When the thief steals from rich and poor equally.
- d. When the thief steals from each of unique houses (living on streets, very poor, poor, moderate, rich, very rich).

2. Dependency Tree (0.5 points)

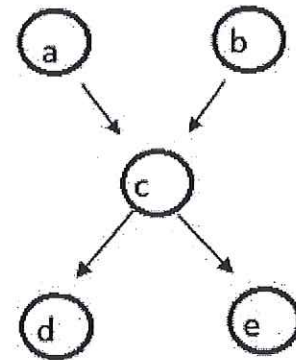
Among the relationships given below which is true

1. $d \perp\!\!\!\perp e \mid c$

2. $a \not\perp\!\!\!\perp b \mid \Phi$

3. $b \not\perp\!\!\!\perp d \mid c$

4. $a \not\perp\!\!\!\perp b \mid c$



a. Option 1 and 4 are true

b. Option 2 and 3 are true

c. Option 1 only true

d. Option 3 and 4 are true

3. The closer you get, the more is rewarded (0.5 point)

There are 10 states as mentioned below. Where ever you may be but you want to reach the final state. The reward of reaching the final state is 1000. Fill the maximum reward of all the states. All the states follow discounted reward with discount factor of 0.9.

A	B	C	Final State
D	E	F	G
H	I		

3.b. Think it Through (0.5 point)

Differentiate Stream based active learning from pool based active learning with neat diagram.

4. Zoologist confused with Tiger and Cheetah (2 points)

Assume you are a zoologist. You are very familiar in identifying the cubs of Tiger and Cheetah. You would have seen a number of cubs of these wild animals and would have studied the different attributes that differentiates them. Finally you build a decision boundary to separate both the Tiger and Cheetah cubs based on their attributes in n-D space. One fine day, you happen to come across a small cub and yet you are unable to identify it. How do you predict whether it's a Tiger or Cheetah cub?

Identify the implied methodology to predict and derive the formulae that uniquely identifies any cub that you encounter in future as well. Draw the Geometric representation.

5. Stop before it's too high (2 points)

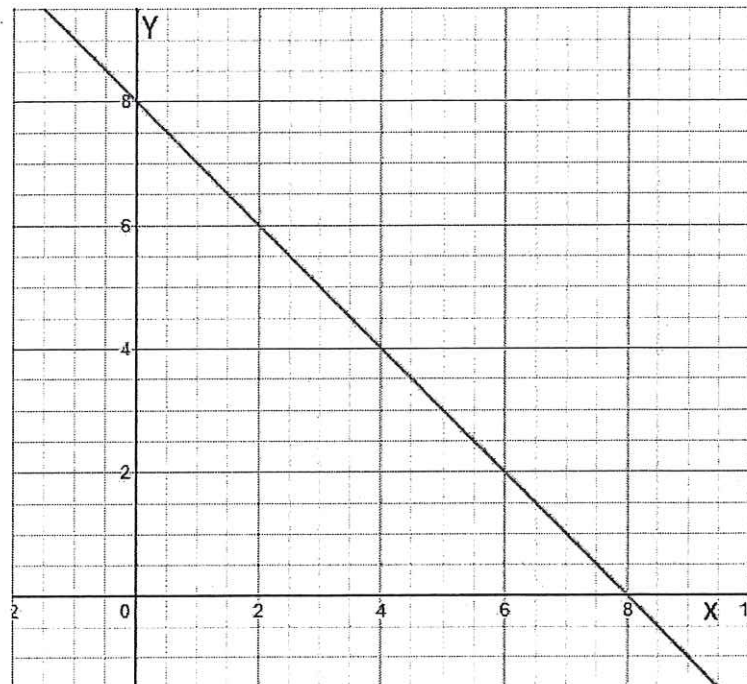
Sketched in the graph below is the derived regression equation. Plot the following points.

Point 1: (2,6)

Point 2: (3,5)

Point 3: (4,4)

Point 4: (6,2)



1. Pen down the Error function formulae and calculate the total error value with the 4 points mentioned above. Do you think the regression equation is the idle equation?

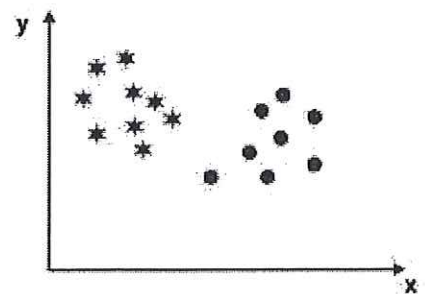
2. Pen down the Error function with regularization parameter. Calculate the new error values by taking regularization value as 0, 1 and 2. Justify the different values of regularization parameter with error values.

6. Let it maximize (4 points)

Understand the diagram. Identify the method that separates the two classes far apart with a hyperplane

Derive the formulae to find the maximum width of the hyperplane separating the two class of data points.

How do you maximize the width of the street if the points are not overlapping with each other? What factors does it depend upon??



7. Drive into and not out (4 points)

A building has two gates (Gate X and Gate Y) both leading to same parking lot. It was found that 90 out of 100 vehicles use Gate X and 20 out of 100 vehicles use Gate Y to enter into campus.

In a period of 10 minutes:

1. When both the gates let the vehicles in, then 80% of the parking lot will be filled.
2. When both the gates does not let the vehicle in then of course, no vehicles will be in parking lot.
3. When only Gate X is left open, 40% of the parking lot will be filled.
4. When only Gate Y is left open, 20% of the parking lot will be filled.

Remember both share the same parking lot. Unfortunately, today the parking lot is empty. It may be because no vehicles have entered into the campus.

You identify the posterior probability of (parking lot being empty / no vehicles have entered the campus via Gate X).

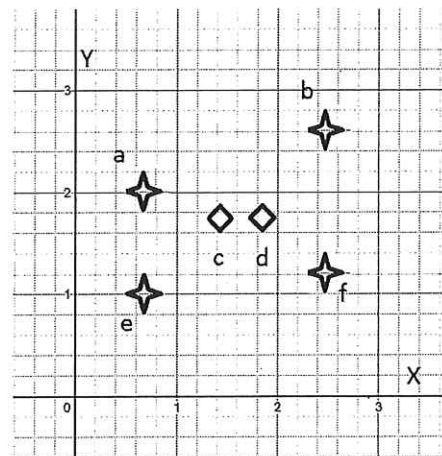
Also find the maximum likelihood of parking lot being empty because no vehicles have entered the campus via Gate X.

What do you infer from both the results derived??

8. Build a strong house with small weak bricks (4 points)

Given the following points from two different class of data. Build a strong classifier by constructing a number of weak classifiers such that it classifies both the class of points approximately.

Finally the equation of the strong classifier.



9. a. Melange of Models [3 point]

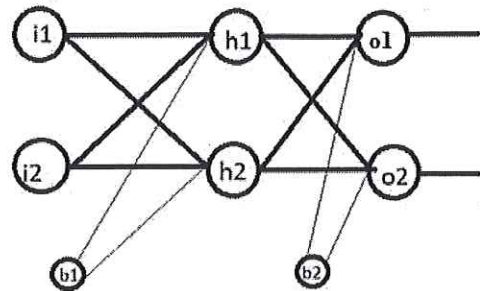
Customize the Bayes rule and derive the maximal posterior probability equation of a data set by using any soft clustering approach that you are familiar with.

9.b. Burglary Alarm [1 point]

Draw the truth table of McCulloch-Pits neuron with three input OR gate and draw the geometric representation signifying the same.

10. Fine tuning the network but how long??? (9 points)

Six neurons are arranged in 2 layers. With two neurons in input layer, two in hidden layer and two in output layer. Assume the activation function to be sigmoid unit.



Input at $i1 = 0.05$, Input at $i2 = 0.10$

Weight ($i1, h1$) = 0.1, Weight ($i1, h2$) = 0.2, Weight ($i2, h1$) = 0.2, Weight ($i2, h2$) = 0.3

Weight ($h1, o1$) = 0.4, Weight ($h1, o2$) = 0.4, Weight ($h2, o1$) = 0.5, Weight ($h2, o2$) = 0.5

$b1 = 0.3$, $b2 = 0.6$, learning rate = 0.5, Expected Target values at $o1$ and $o2$ is 1 respectively.

1. Derive the formulae for total error propagated via link $i1-h1$ and link $h1-o1$ (Using Partial Derivatives). (6 points)
2. With the help of the formulae that you derived in section 9.b.1, calculate the total error of the entire circuit. (2 point)
3. Calculate the error propagated via link $h1-o1$ and find the new weight of link $h1-o1$ link. (1 point)

11. Too much math involved to reduce the weight of anything (9 points)

Given the data set apply a supervised dimensionality reduction approach and find the new values of all the data points using linear discriminant function. (7 points)

Is there any change in points allocated to its respective class label? (1 point)

Can you further reduce the dimension of the dataset? (1 point)

x	y	CLASS
2	1	1
2	2	1
2	3	1
4	5	2
4	4	2
4	3	2

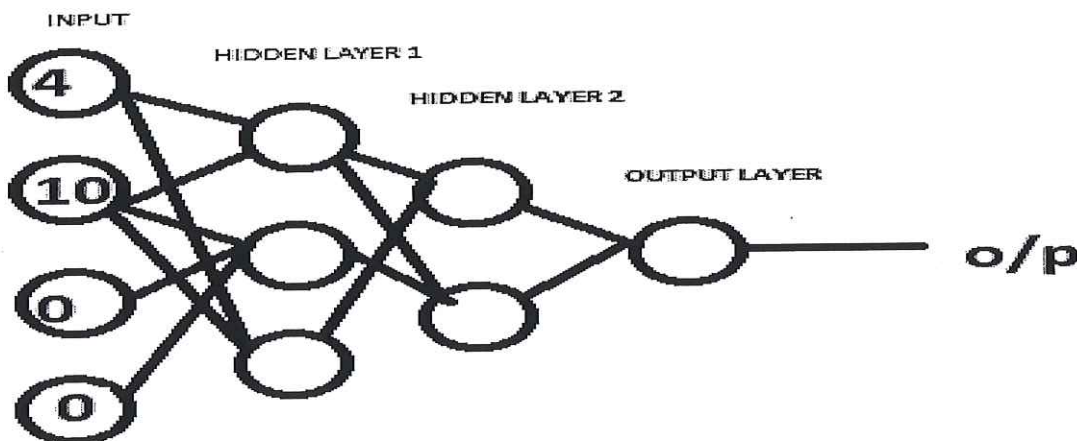
Quiz

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ACADEMIC-UNDERGRADUATE STUDIES DIVISION
BITS F464- MACHINE LEARNING
FIRST SEMESTER 2019-2010

TIME : 20 MINUTES
DATE: 14.10.2019

MARKS : 10

1. Write is the derivate of sigmoid function, $y(x)$. (1 point)
2. What is a perceptron? (1 point)
 - a. A multi-layer network of neuron
 - b. A single layer network of neuron
 - c. A single layer network of neuron with feed forward circuit
 - d. A neuron whose input link is not associated with weight.
3. Given the neural network below:
Assume that the nodes in hidden layer1 perform Boolean AND operation; the nodes in hidden layer2 and output layer performs Boolean OR operation, calculate the output of the network. (2 point)



4. What problem is caused when the learning rate is too high during gradient calculation? (1 point)
 - a. Cost function does not converge to an optimal solution and can even diverge.
 - b. Cost function may not converge to an optimal solution, or will converge after a very long time.
 - c. Cost function oscillates.
 - d. Will surely end in global minima.
5. When the learning process in backpropagation is to be stopped. (1 point)
 - a. When the cost function reduces and gradient convergence involved
 - b. No special heuristic criteria need to be fixed
 - c. When the cost function reduces and the average gradient convergence involved
 - d. none of the above

6. Assume you have 5 learners .

(1 point)

Learner 1's prediction is 90% right

Learner 2's prediction is 90% wrong

Learner 3's prediction is 50% right and 50% wrong

Learner 4's prediction is 60% right and 40% wrong

Learner 5's prediction is 40% right and 60% wrong

Which combination of learners would wisely help in decision making process?

1. Learner 1 and Learner 5

2. Learner 2 and Learner 4

3. Learner 3 ,4, and 5

4. Learner 1 and 2

7. Assuming you to be a decision tree learner, given a matrix $X = \begin{vmatrix} 6 & 3 \\ 2 & 7 \\ 9 & 6 \\ 4 & 2 \end{vmatrix}$ and the labels be $Y = \begin{vmatrix} 1 \\ 0 \\ 1 \\ 0 \end{vmatrix}$

Let x_1 denote the first feature in matrix X and x_2 denote the second feature. Identify the best split using information gain

a. $x_1 > 2$

b. $x_2 > 3$

c. $x_1 > 4$

d. $x_2 > 6$

NOTE: LOGICALLY THINK, BEFORE YOU START SOLVING....

(2 points)

8. Select the statements that are true over Gradient descent

(1 point)

a. The starting point of gradient descent will decide reaching the global minima

b. The slope of gradient descent will decide reaching the global minima

c. Plotting the gradient descent over contour graph, the direction of gradient will move from inner to outer rings

d. Plotting the gradient descent over contour graph, the direction of gradient will move from outer to inner rings

PEN DOWN YOUR ANSWERS

1	
2	
3	
4	
5	
6	
7	
8	